Solution

i) The threshold wavelength is given by

$$\lambda_0 = \frac{hc}{\phi_0} = \frac{6.626\times10^{-34}\times3\times10^{8}}{4.65\times1.6\times10^{-19}}$$

$$= 2672\ \text{Å}$$

ii) Energy of the photon of wavelength 2200 Å is

$$E = \frac{hc}{\lambda} = \frac{6.626\times10^{-34}\times3\times10^{8}}{2200\times10^{-10}}$$

$$= 9.035\times10^{-19}\,J = 5.65\,\text{eV}$$

We know that kinetic energy of fastest photo electron is

$$K_{max} = h\nu - \phi_0 = 5.65 - 4.65$$

$$= 1\ \text{eV}$$

From equation (8.3), $K_{max} = eV_0$

$$V_0 = \frac{K_{max}}{e} = \frac{1\times1.6\times10^{-19}}{1.6\times10^{-19}}$$

Therefore, stopping potential = 1V

EXAMPLE 8.4

The work function of potassium is 2.30 eV. UV light of wavelength 3000 Å and intensity 2 Wm^{-2} is incident on the potassium surface. i) Determine the maximum kinetic energy of the photo electrons ii) If 40% of incident photons produce photo electrons, how many electrons are emitted per second if the area of the potassium surface is 2 cm^2 ?

Solution

i) The energy of the incident photon is

$$E = \frac{hc}{\lambda} = \frac{6.626\times10^{-34}\times3\times10^{8}}{3000\times10^{-10}}$$

$$E = 6.626\times10^{-19}\,\text{J} = 4.14\ \text{eV}$$

Maximum KE of the photoelectrons is

$$K_{max} = h\nu - \phi_0 = 4.14 - 2.30 = 1.84\ \text{eV}$$

ii) The number of photons reaching the surface per second is

$$n_p = \frac{I}{E}\times A$$

$$= \frac{2}{6.626\times10^{-19}}\times2\times10^{-4}$$

$$= 6.04\times10^{14}\ \text{photons / sec}$$

The rate of emission of photoelectrons is

$$= (0.40)n_p = 0.4\times6.04\times10^{14}$$

$$= 2.416\times10^{14}\ \text{photoelectrons / sec}$$

EXAMPLE 8.5

Light of wavelength 390 nm is directed at a metal electrode. To find the energy of electrons ejected, an opposing potential difference is established between it and another electrode. The current of photoelectrons from one to the other is stopped completely when the potential difference is 1.10 V. Determine i) the work function of the metal and ii) the maximum wavelength of light that can eject electrons from this metal.

Solution

i) The work function is given by

$$\phi_0 = h\nu - K_{max} = \frac{hc}{\lambda} - eV_0$$

since $K_{max} = eV_0$

$$= \left[\frac{6.626\times10^{-34}\times3\times10^{8}}{390\times10^{-9}}\right] - \left[1.6\times10^{-19}\times1.10\right]$$

$$= 5.10\times10^{-19} - 1.76\times10^{-19} = 3.34\times10^{-19}\,\text{J}$$

$$= 2.09\ \text{eV}$$

ii) The threshold wavelength is

$$\lambda_0 = \frac{hc}{\phi_0} = \frac{6.626\times10^{-34}\times3\times10^{8}}{3.34\times10^{-19}}$$

$$= 5.951\times10^{-7}\,\text{m} = 5951\ \text{Å}$$

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